ASSIGNMENT 2 EVIDENCE FILE: OPERATIONAL MODEL and EVIDENCE FOLDER

Course: Business Administration (BA2026) | Focal Company: JSC NC 'KazMunayGas' (KMG) | Comparative: Shell plc

Student Name: Kenzhebay Ayaulym
Group: DPA-2602
Components Included:
Week 4 Human-AI Matrix
Week 5 Process Map and Controls
Running Evidence Folder
AI-Use Declaration

1. Week 4 Human-AI Task Allocation Matrix (KMG)

The matrix below evaluates eight core operational tasks across KazMunayGas's POLC management framework (Planning, Organizing, Leading, Controlling), categorizing each into one of four Human-AI operating modes based on rigorous test-sentence justifications.

#	Function	KMG Operational Task	Category	Justification (Test + KMG Fact)	Accountable Role	Review Point
1	Planning	Allocate annual CAPEX across regional assets	Human + AI	Cost of Error Test: Strategic capital decisions impact 0.16x leverage; AI models scenarios, CFO decides.	CFO	No
2	Planning	Forecast domestic fuel demand for ANPZ/PNHZ	AI Executes + Human Review	Volume & Rules Test: Algorithmic demand forecasting across 14 regions; Head of Marketing verifies.	Head of Marketing	Yes
3	Organizing	Workforce restructuring in Mangystau assets	Human Only	Accountability & Empathy Test: Affects >52,000 workers; social stability mandates strict human management.	HR Director	No

4	Organizing	Schedule routine refinery valve/pump inspections	Automated	Volume & Rules Test: Fixed sensor thresholds trigger automated work tickets; manual fallback log available.	Refinery Lead	No
5	Leading	Executive performance and dismissal reviews	Human Only	Accountability Test: Fiduciary and legal employment rights under KZ Labour Code required.	Board / MD	No
6	Leading	Monitor workplace safety breaches on workovers	AI Executes + Human Review	Cost of Error Test: TUMAR computer vision flags PPE breaches; Chief HSE Inspector verifies.	Chief HSE Inspector	Yes
7	Controlling	Predictive failure monitoring at refineries	AI Executes + Human Review	Cost of Error Test: Prevents unplanned downtime; Chief Mechanical Engineer signs off diagnostics.	Chief Engineer	Yes
8	Controlling	Waterflooding rate adjustment across 1,660 cells	AI Executes + Human Review	Volume & Rules Test: ABAI AI optimizes sweep; Chief Reservoir Engineer approves pressure changes.	Reservoir Engineer	Yes

2. Week 5 Process Mapping & Operational Control Framework

Process Selected: Refining Equipment Maintenance Workflow at Atyrau Refinery (ANPZ).
Key Performance Indicator (KPI): Reduce unplanned refinery downtime by 15% (baseline: 120 hrs/yr) and lower maintenance OPEX by 10% by end of 2027.

A. Current-State Process Map (Manual Baseline)

#	Actor (Swimlane)	Step (Verb + Object)	Time / Delay	Bottleneck Identified
1	Field Operator	Perform physical inspection walk-throughs	2 hrs / shift	No
2	Local Gauge	Record temperature and	1 hr	No

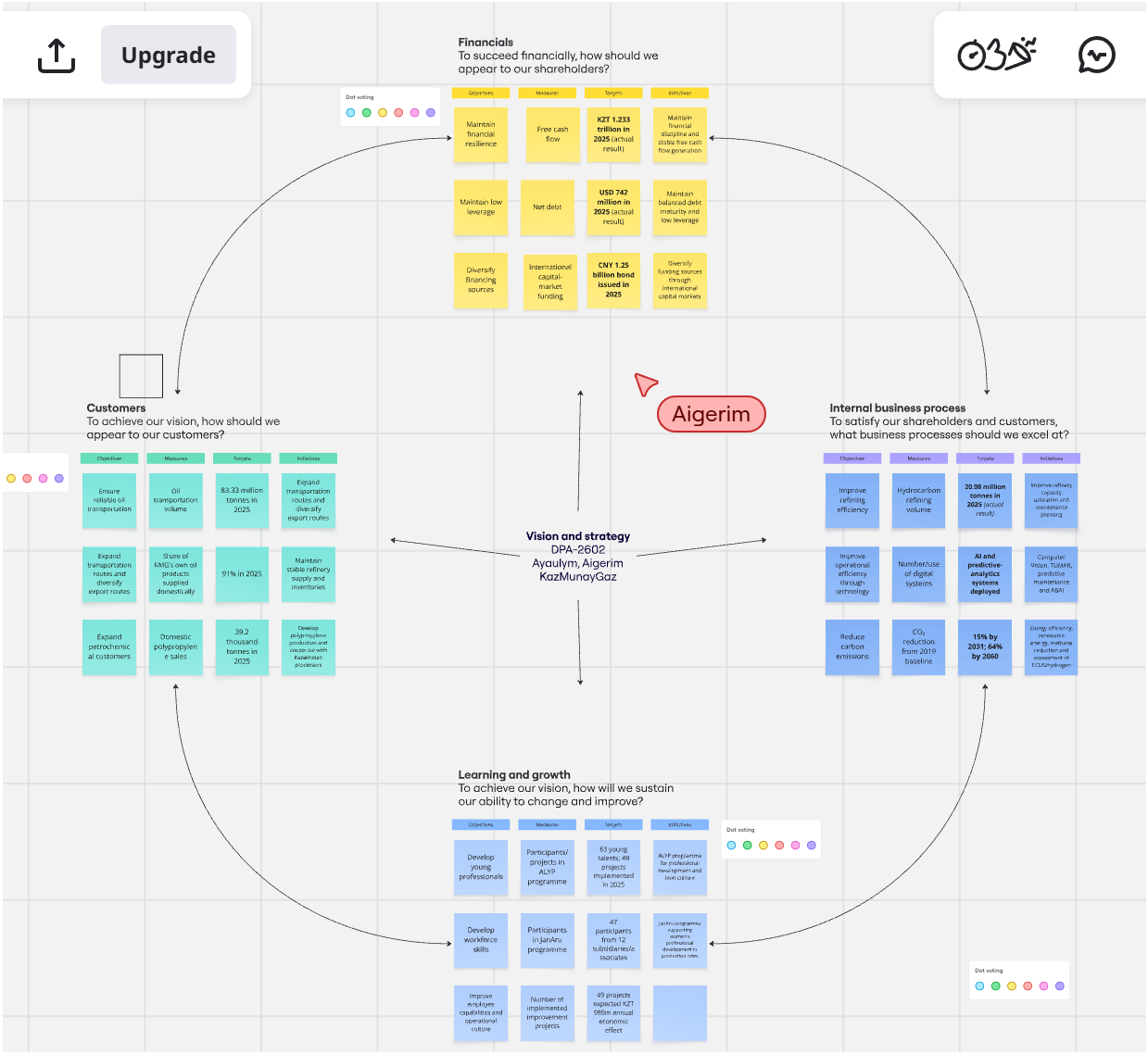
		vibration figures		
3	Maintenance Staff	Analyze historical failure trends in Excel	12-24 hrs	BOTTLENECK (reactive)
4	Chief Engineer	Approve emergency repair work orders	4 hrs	No
5	Repair Crew	Shut down unit and execute repairs	48-72 hrs	No (Downtime accrued)

B. Future-State Process Map (AI-Integrated Controls)

#	Actor / System	Step (Verb + Object)	Category	What Changed & Impact on KPI
1	IoT Telemetry	Stream real-time vibration/thermal telemetry	Automated	Replaced physical walk-throughs; continuous feed.
2	Predictive AI	Process telemetry and flag failure anomalies	AI Executes + Review	Eliminates bottleneck; predicts failure 14 days ahead.
3	Chief Engineer	Verifying anomaly flag and approve work order	Human + AI	Human Review Point: Prevents false shutdowns.
4	Automated System	Schedule parts procurement and crew dispatch	Automated	Cuts planning lead time from 24 hrs to 30 mins.
5	Maintenance Crew	Execute planned preventative maintenance	Human Only	Achieves target 15% downtime reduction KPI.

C. Operational Control Elements

• **Human Review Point:** Certified Chief Mechanical Engineer must independently review AI failure probability logs and sign off before initiating any physical plant shutdown.
• **Escalation Path:** AI Anomaly Flag, Senior Tech (15 mins), Chief Mechanical Engineer (1 hr) , General Director (4 hrs if repair >24 hrs).
• **Manual Fallback Plan:** If IoT sensors or AI servers fail, plant operators immediately revert to manual hourly operator logbooks, analog pressure gauge walks, and standard calendar maintenance.



3. Running Evidence Folder (IEEE Reference List, Current to Week 5)

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. Houston, TX, USA: OpenStax, Rice University, 2024.

[2] JSC NC 'KazMunayGas', Annual Report 2025: Global Standards. National Priorities, Astana, Kazakhstan, Apr. 28, 2026.

[3] Shell plc, Annual Report and Accounts 2025: Deliver More Value with Less Emissions, London, UK, Mar. 11, 2026.

[4] JSC NC 'KazMunayGas', Consolidated Financial Statements for the Year Ended December 31, 2025, PwC Audit Report, Astana, Mar. 5, 2026.

[5] KMG Engineering, ABAI System Operational Brief and Digitalization Roadmap, Astana, Kazakhstan, 2026.

[6] KAZENERGY Association, National Energy Report 2023, Astana, Kazakhstan, 2023.

[7] Wendover Productions, How Boeing Lost Its Way, YouTube, 2024. [Online]. Available: https://www.youtube.com/watch?v=URoVKPVDKPU

[8] Klarna, Klarna AI Assistant Operational Release & Quality Calibration Review, Press Release, Feb. 2024.

[9] Ministry of Energy of the Republic of Kazakhstan, Digitalization of Energy Sector and Refining Outlook, Official Report, Astana, 2026.

[10] Caspian Pipeline Consortium (CPC), Operational Crude Oil Transportation Volumes Brief, Almaty, Kazakhstan, 2026.

4. Completed AI-Use Declaration Form (Level 2: Supported & Disclosed)

Policy Level Selected: Level 2 - Supported and Disclosed (AI used for brainstorming, structural organization, and citation formatting with mandatory independent factual verification).

Table C: Detailed AI Activity Log

AI System Used	Specific Task & Prompts	How Output Was Verified	Changes Made by Student
ChatGPT	Grammar and error checking	Checked grammar, spelling, and wording of the text	All ideas, analysis, and final content were created and reviewed by the student
Gemini	Formatting IEEE references and generating test-sentence templates for Task Matrix.	Verified OpenStax Chapter 6-7 citations and KZ Labour Code legal accountability.	Rewrote justifications to ensure KMG social mandate was explicitly emphasized.
ChatGPT	Translation	Translated short phrases or sentences when needed	Student checked and adapted the translations to the contex

Section F: Reflection on Rejected AI Suggestions

AI originally suggested copying Shell's strategy of divesting low-margin refining assets and floating domestic fuel prices. This suggestion was REJECTED because KazMunayGas holds a sovereign obligation to maintain domestic energy security and protect employment across regional refineries. Automated workforce reduction suggestions were also rejected due to Mangystau regional labour agreements.

Section G: Final Integrity Confirmations & Signature

☑ I confirm that the submitted script (400–500 words) and analysis represent my own independent work.

☑ I confirm that all financial figures and operational facts have been independently verified.

☑ I confirm that all AI interactions are fully disclosed above in accordance with Level 2 policy.

Student Signature: Ayaulym Kenzhebay **Date:** October 4, 2026